Data Migration Plan

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Data Migration Objective

The primary objective of this data migration project is to seamlessly transition the clients organizational data assets from legacy systems to Salesforce CRM and CPQ records. By prioritizing data quality, integrity, and security, while minimizing operational downtime, this project aims to consolidate disparate data sources into a unified system. Ultimately, this migration will reduce infrastructure overhead, and empower the organization with enhanced data accessibility and operational efficiency.

In Scope / Out of Scope

The following table outlines what source systems and relevant record types will be in scope along with what remains out of scope for the resources on this data migration project:

Entity / Data Category	In Scope	Out of Scope	Notes / Justification	Client Signature
Core CRM Data	Active Accounts (has an active contract) Contacts - at least the three most recent contacts created for their related Account	- In-Active Accounts (Accounts without an active contract) - Contacts related to inactive Accounts - Open Leads - Open Opportunities - Historic email and any notes	Historical Accounts and related Contacts that no longer have an active contract will be archived in a data warehouse.	

		related to contacts		
CPQ Product Catalog	None (Product catalog, pricing rules, and discount schedules will be **rebuilt natively** in the new org)	- Legacy product catalog rules from source system - Hardcoded or deprecated price books and legacy bundles	Prevents sales reps from quoting old, unsupported products in the new system.	
CPQ Quotes and Quote Lines	None (only completed quotes that have been processed into Contracts and are active will migrated using data from contract and contract line tables	Open Quotes with Quote Lines that are not converted to active contracts will remain in source system or be stored in Data Warehouse		
Contracts, Assets, and Subscriptions	- Active Contracts that have a start date before today and end date after today that are for 1 year. - Assets and subscriptions related to active contracts will be migrated	- Active Contracts that are multi year contracts. No MDQ will be created in this migration.	Only single year active contracts will be migrated and all legacy contracts with related subscriptions and assets will be stored in a data warehouse.	
Billing records	None (no billing data will be included in this migration, specifics are listed in Out of Scope	All the record types for billing contract and subscription data below are out of scope: - Invoice - Invoice Line - Credit Memo - Debit Memo - Payment - Payment Allocation	No Billing data will be included in this migration	

Apex Development	None	All custom Apex development will be handled and deployed by the Client's assigned lead Apex SME	All custom Apex will be built and deployed by Client's Apex development team	
Data warehousing historic data	None	All historical data that is stored in Data Warehouses will be handed by the client's team		
Client homegrown systems	Only source data will be migrated from MySQL to Salesforce.	No data will be migrated back into client's MySQL system from Salesforce	Any circular migration effort from salesforce back to client's original system will be handled by the client	
Duplicates	All duplicate records will be reported back to the client for review before migrating any.	All duplicates will be out of scope.		
Custom Fields	- Custom fields will be added to all migrated objects: - Migrated Record: indicates the record is migrated or generated in system by a user - Legacy ID: Used to identify the source record to perform QA on migration effort	Any Custom fields not stated here will be out of scope for this migration effort.	These custom fields are necessary to ensure data migration quality and guarantee records are migrated and related correctly in the target Salesforce system. The fields will be deployed from	

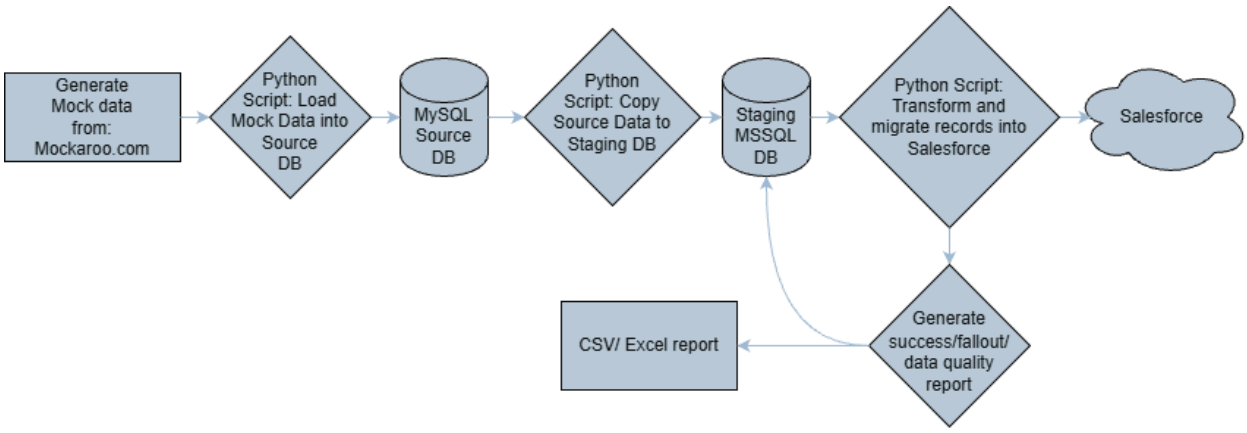
Migration Process Overview

The overall process of this migration will be through an (ELTL) Extract - Load - Transform - Load process. In Scope data will be migrated from the source MySQL system as is to an intermediary staging MSSQL database set up to mirror the source system. This will provide an immutable data source so the client may continue using their source system as they need. This will prevent any changes in the source data from affecting results during script development and Quality Assurance testing. From the staging db, records will be transformed and migrated into Salesforce with the sequence of steps outlined in the Data Migration Runbook document. The Data Migration Runbook will also contain backout scripts used to delete any and all migrated records during testing and be available in the event of issues in production that require us to postpone the production migration.

The Migration will have 4 target systems throughout the project:

- **DEV:** A smaller development environment for apex and system integration engineers to develop their code and deploy up to SIT for testing. Only a small subset of each record will be migrated into this Salesforce org due to storage size constraints, typically a 5MB org which can only store around 2500 records total.
- **SIT**: Systems Integration Testing environment for script development and ensuring data that flows downstream to outside systems works correctly.
- **UAT**: Once the majority of errors and bugs have been resolved and scripts are finished, A User Acceptance Testing environment will be set up to practice for a full production migration. Depending on the first iteration of UAT, successive migration efforts to test for a production run may be necessary. This may include reducing the scope of records migrated in case of source data issues, or dividing the volume of data into multiple tranches to migrate.
- **Production**: Once the client has accepted the results of the latest UAT migration as sufficient the lead data engineer and client will confirm any planned production run date.

Below is a graph illustrating the migration process:



Reporting

The reporting process in this migration will include the following:

- Number of Records Processed
    - Number of records successfully processed
    - Number of records that fall out due to errors in either source data or system error.
- Processing time of each step to identify bottlenecks and estimate overall data migration time. This will help determine how long the production target system needs to be down for.
    - Transformation timing of records
    - Upload timing of records
    - Total script time
- Success and Fallout reporting
    - Reports can be created in any of the three following options
        - CSV reports
        - Excel reports
        - Loading reports back into staging db
    - Each migration script will produce a success and fallout report for the final upload
    - Fallout reports will state reason for failure
        - Salesforce will provide reason for upload failure
        - Data Engineer will write reason for failure of any records not uploaded
    - These reports will be independent from any Quality Assurance testing and reporting.

Data Validation

The Data Validation process in this migration will include the following:

- Source to Staging Data Validation
    - After the source records are loaded into the MSSQL staging environment a data validation test will be run for each individual record type.
    - These tests will verify:
        - the data type of each column
        - the data value in each column per record
        - The number of records staged per record type
- Staging to Salesforce Data Validation
    - After each migration script inserts a new record or updates existing records in Salesforce a test will be run to verify before proceeding to the next object type.
    - These tests will verify:
        - the data type of each field
        - the data value in each field per object type
        - The number of records migrated into each object type

Roles & Responsibilities

- Lead Data Migration Engineer
    - Create Deliverable Documents:
        - Design Data Migration Runbook
        - Design Data Migration Mapping Document
        - Design Data Migration Plan Overview
    - Create Data Migration Scripts.
    - Set up a Data Migration Staging database.
    - Generate reports and meet with client SMEs to understand how to interpret reports.
    - Assigned System Administrator role permissions in Salesforce.
        - This is generally necessary for backing out records, which is extensively done during script development phase in the SIT environment.
    - Create a Data Migration backout process to purge the system of any migrated records.
        - Steps will be outlined in the Data Migration Runbook document.
        - These backout scripts are developed in tandem with migration scripts.
- Data Migration Engineer
    - Create Data Migration Scripts.
    - Set up a Data Migration Staging database.
    - Set up Fallout reporting for both data issues and upload issues.
    - Assigned System Administrator role permissions in Salesforce.
        - This is generally necessary for backing out records, which is extensively done during script development phase in the SIT environment.
    - Create a Data Migration backout process to purge the system of any migrated records.
        - Steps will be outlined in the Data Migration Runbook document.
        - These backout scripts are developed in tandem with migration scripts.
- Quality Assurance
    - Create tests to assure the migrated data functions as expected.
    - Create reports that the source data records match the migrated records 1-to-1 according to the Data Migration Mapping document.
    - Report any duplicate records or fields that are incorrectly mapped in the migration scripts according to the Data Migration Mapping document.
- Client CRM Data SME
    - Aid in Data Migration Mapping document for Accounts and Contacts
    - Provide aid and guidance to Data Engineer in resolving source data issues/inconsistency
- Client Contract Data SME
    - Aid in Data Migration Mapping document for Opportunities, Quote, Quote Line, Order, OrderItem, Contract, and Subscription.
    - Provide aid and guidance to Data Engineer in resolving source data issues/inconsistency

- Client Lead Apex Developer
    - If certain access restrictions arise in regards to the Data Engineers creating custom fields and accessing the installed package to turn on/off CPQ triggers. A dedicated resource is necessary to aid in these activities.
    - Any field overrides that are necessary to be built for the migration will be handled by the Client's Apex developers

Tools & Software

- Staging DB - The staging database will be a MSSQL database with full write/read access provided to the data engineering team.
- Migration and reporting during the ELTL process will be handled by custom python scripts built around an inhouse data processing framework.